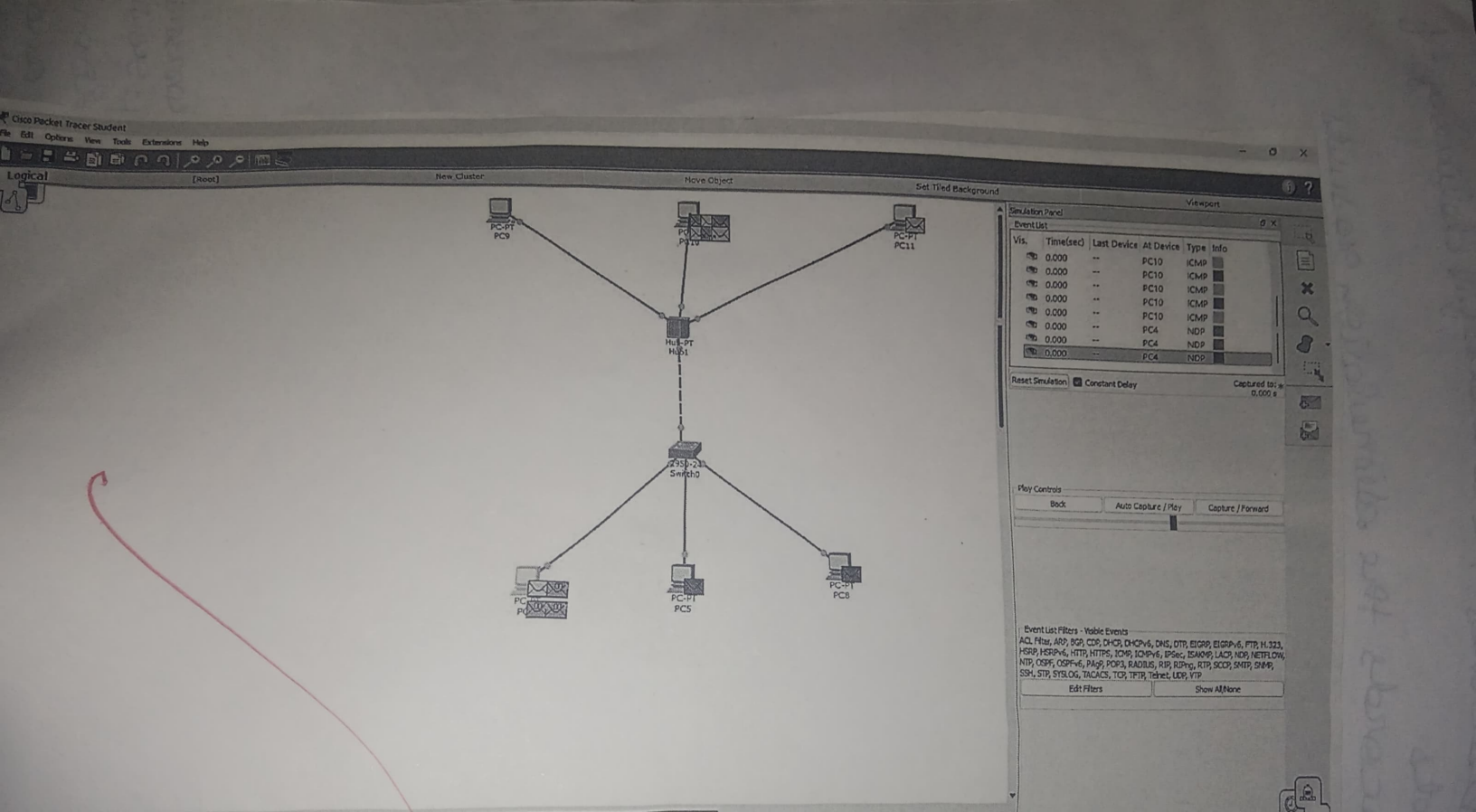
ii) WHEN SUBNETS ARE NOT CONNECTED THROUGH A ROUTER

AIM:

To analyze packet transmissions between different subnets when no router is available.

STEPS:-

1. Remove the router from the network topology
2. Open simulation mode
3. Select devices belonging to different subnets
4. Generates packets using the simple PDU tools
5. Observe the packet transmission
6. Perform a Ping test between devices located in different subnets
7. Analyze the packet flow
8. Observe the simulation events
9. Identify the points at which Packet transmissions fails
10. Compare the results with the networks that uses a router
11. Record the Observation
12. Analyze the importance of a router in inter-network communication


Cisco Packet Tracer Student
File Edit Options View Tools Extensions Help
Logical
[Root]
New Cluster
Move Object
Set Tiled Background
Viewport
Simulation Panel
Event List
Vis. Time(sec) Last Device At Device Type Info
0.000 -- PC10 ICMP
0.000 -- PC10 ICMP
0.000 -- PC10 ICMP
0.000 -- PC10 ICMP
0.000 -- PC10 ICMP
0.000 -- PC4 NDP
0.000 -- PC4 NDP
0.000 -- PC4 NDP
Reset Simulation
Constant Delay
Captured to:
0.000 s
Play Controls
Back
Auto Capture / Play
Capture / Forward
Event List Filters - Visible Events
ACL Filter, ARP, BGP, CDP, DHCP, DHCPv6, DNS, DTP, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPSec, ISAKMP, LACP, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, RADIUS, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, VTP
Edit Filters
Show All/None
PC-PT PC9
PC-PT PC11
Hub-PT Hub1
2950-24 Switch0
PC-PT PC5
PC-PT PC8

RESULT:-

Thus, Communication between devices located in different subnets failed when router was not used. The packets could not reach the destination. this exp. demonstrated the importance of routers in enabling communication between different subnetworks

PC0

Physical | Config | Desktop | Custom Interface

IP Configuration

IP Configuration

- ○ DHCP
- ● Static

IP Address	192.168.1.11
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	

IPv6 Configuration

- ○ DHCP
- ○ Auto Config
- ● Static

IPv6 Address	/
Link Local Address	FE80::2D0:BAFF:FE9C:D809
IPv6 Gateway	
IPv6 DNS Server	